

Change in basis

$\underline{v}$ and $\underline{s}$ are invariant

representations $[\underline{v}]$ and $[\underline{s}]$ are not

Two frames $\{\underline{e}_i\}$ and $\{\underline{e}'_i\}$

Representation of $\underline{e}'_j$ in $\{\underline{e}_i\}$

$$\begin{aligned} [\underline{e}'_j] &= (\underline{e}'_j \cdot \underline{e}_1) \underline{e}_1 + (\underline{e}'_j \cdot \underline{e}_2) \underline{e}_2 + (\underline{e}'_j \cdot \underline{e}_3) \underline{e}_3 \\ &= \underbrace{(\underline{e}'_j \cdot \underline{e}_i)}_{A_{ij}} \underline{e}_i \end{aligned}$$

$$[\underline{e}'_j] = A_{ij} \underline{e}_i \quad \text{where} \quad A_{ij} = \underline{e}_i \cdot \underline{e}'_j$$

↑
note transpose

Here $\underline{A}$ is the change of basis tensor

$$\underline{A} = A_{ij} \underline{e}_i \otimes \underline{e}_j \quad A_{ij} = \underline{e}_i \cdot \underline{e}'_j$$

What kind of tensor?

Similarly we express $\underline{e}_i$ in $\{\underline{e}'_k\}$

$$[\underline{e}_i]' = (\underline{e}_i \cdot \underline{e}'_k) \underline{e}'_k = A_{ik} \underline{e}'_k$$

$$\underline{e}'_j = A_{ij} \underline{e}_i$$

$$\underline{e}'_j = A_{ij} (A_{ik} \underline{e}'_k)$$

$$= A_{ij} A_{ik} \underline{e}'_k$$
$$\underbrace{\hspace{1.5cm}}_{\delta_{jk}}$$

$$\underline{e}'_j = \delta_{jk} \underline{e}'_k$$

$$\underline{e}_i = A_{ik} \underline{e}'_k$$

$$\underline{e}_i = A_{ik} (A_{lk} \underline{e}_l)$$

$$\underline{e}_i = \underbrace{A_{ik} A_{lk}}_{\delta_{il}} \underline{e}_l$$

$$\underline{e}_i = \delta_{il} \underline{e}_l$$

$$\Rightarrow A_{ij} A_{ik} = \delta_{ik}$$

$$\Rightarrow A_{ik} A_{lk} = \delta_{il}$$

$$\underline{\underline{A}}^T \underline{\underline{A}} = \underline{\underline{A}} \underline{\underline{A}}^T = \underline{\underline{I}}$$

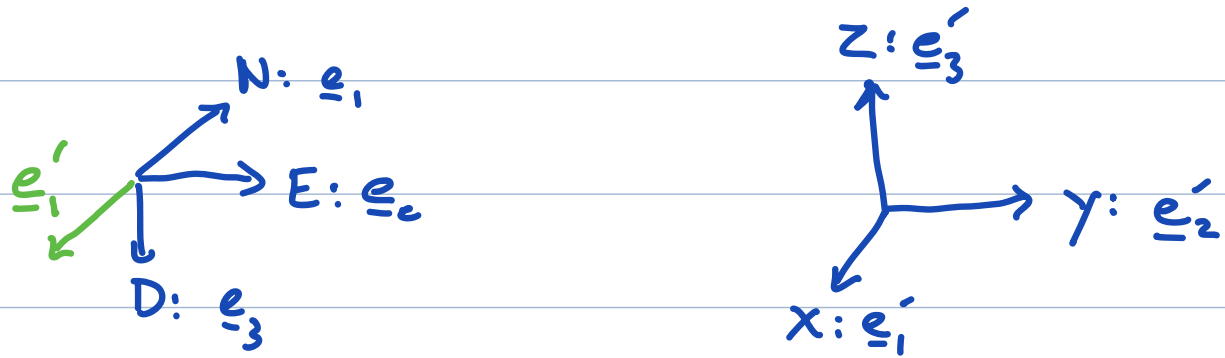
$\Rightarrow \underline{\underline{A}}$ is orthonormal

if $\{\underline{e}_i\}$ and $\{\underline{e}'_i\}$ are right handed

$$(\underline{e}_i \times \underline{e}_j) \cdot \underline{e}_k = \epsilon_{ijk} \quad (\underline{e}'_i \times \underline{e}'_j) \cdot \underline{e}'_k = \epsilon_{ijk}$$

$\Rightarrow \underline{\underline{A}}$ must be rotation $\det(\underline{\underline{A}}) = 1$

Example: Consider NED $\{\underline{e}_i\}$ and XYZ $\{\underline{e}'_i\}$



$$[\underline{e}'_1] = (\underbrace{\underline{e}_1 \cdot \underline{e}'_1}_{-1}) \underline{e}_1 + (\underbrace{\underline{e}_2 \cdot \underline{e}'_1}_0) \underline{e}_2 + (\underbrace{\underline{e}_3 \cdot \underline{e}'_1}_0) \underline{e}_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$[\underline{e}'_2] = [\underline{e}_2] = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad [\underline{e}'_3] = -[\underline{e}_3] = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$\underline{\underline{A}} = A_{ij} \underline{e}_i \otimes \underline{e}_j$$

$$A_{ij} = \underline{e}_i \cdot \underline{e}'_j$$

$$\underline{\underline{A}} = ([\underline{e}'_1], [\underline{e}'_2], [\underline{e}'_3]) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$\uparrow \quad \uparrow \quad \uparrow$
 $[\underline{e}'_1] \quad [\underline{e}'_2] \quad [\underline{e}'_3]$

Columns of $\underline{\underline{A}}$ are representation of $\underline{e}'_i$ in $\{\underline{e}_i\}$!

Note how this is analogous to stress:

$$\underline{\underline{\sigma}} = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \underline{t}(\underline{e}_1) & \underline{t}(\underline{e}_2) & \underline{t}(\underline{e}_3) \\ \downarrow & \downarrow & \downarrow \end{bmatrix}$$

tractions on coord. planes

traction on any plane:

$$\underline{t}_n = \underline{\underline{\sigma}} \underline{n} = \underbrace{n_1 \underline{t}(\underline{e}_1) + n_2 \underline{t}(\underline{e}_2) + n_3 \underline{t}(\underline{e}_3)}_{\text{linear combination.}}$$

Similarly:

$$\underline{\underline{A}} = \begin{bmatrix} | & | & | \\ [\underline{e}'_1] & [\underline{e}'_2] & [\underline{e}'_3] \\ | & | & | \end{bmatrix}$$

representation of base vectors

representation of any vector.

$$\underline{v} = \underline{\underline{A}} \underline{v}' = \underbrace{v'_1 [\underline{e}'_1] + v'_2 [\underline{e}'_2] + v'_3 [\underline{e}'_3]}_{\text{linear combination}}$$

Change in representation of vectors

$[\underline{v}]$ and $[\underline{S}]$ in $\{\underline{e}_i\}$

$[\underline{v}]'$ and $[\underline{S}]'$ in $\{\underline{e}'_i\}$

$$[\underline{v}] = [\underline{A}] [\underline{v}]' \quad \text{and} \quad [\underline{v}]' = [\underline{A}]^T [\underline{v}]$$

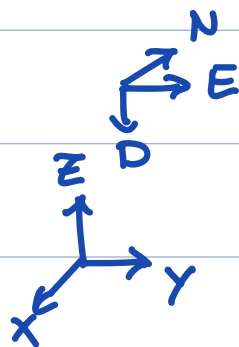
to see this $\underline{v} = v_i \underline{e}_i = v'_j \underline{e}'_j$ but $\underline{e}'_j = A_{ij} \underline{e}_i$
 $= v'_j (A_{ij} \underline{e}_i)$

$$v_i \underline{e}_i = A_{ij} v'_j \underline{e}_i$$

by comparison: $v_i = A_{ij} v'_j$ ✓

Example: NED $\{\underline{e}_i\}$

XYZ $\{\underline{e}'_i\}$



$$\text{HW 1: } [\underline{a}] = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad [\underline{a}]' = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} \quad \text{and} \quad [\underline{A}] = \begin{bmatrix} -1 & & \\ & 1 & \\ & & -1 \end{bmatrix}$$

$$[\underline{A}] [\underline{a}]' = \begin{bmatrix} -1 & & \\ & 1 & \\ & & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = [\underline{a}] \quad \checkmark$$

Change in representation for tensors

$$\begin{aligned} [\underline{S}] &= [\underline{A}][\underline{S}][\underline{A}]^T \\ [\underline{S}]' &= [\underline{A}]^T[\underline{S}][\underline{A}] \end{aligned}$$

$\Rightarrow$ HW 4

with $\underline{A} = A_{ij} \underline{e}_i \otimes \underline{e}_j$ $A_{ij} = \underline{e}_i \cdot \underline{e}'_j$
 $\underline{A}^T \underline{A} = \underline{A} \underline{A}^T = \underline{I}$ $\det(\underline{A}) = 1$ (rotation)
 $\underline{A}^T = \underline{A}^{-1}$

Look familiar? $[\underline{S}]' = [\underline{A}]^{-1}[\underline{S}][\underline{A}]$

Diagonalizing a matrix: $\underline{A} = \underline{V}^{-1} \underline{\Lambda} \underline{V}$

$$\underline{A} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \quad \underline{V} = \begin{bmatrix} | & | & | \\ \underline{v}_1 & \underline{v}_2 & \underline{v}_3 \\ | & | & | \end{bmatrix}$$

$$(\underline{A} - \lambda_p \underline{I}) \underline{v}_p = 0$$

$\Rightarrow$ eigenvalues & eigenvectors

Eigenvalues & vectors of tensors

By the eigen pair of $\underline{\underline{S}} \in \mathcal{V}^2$ we mean the scalar λ and the vector $\underline{v}$ such that

$$\underline{\underline{S}} \underline{v} = \lambda \underline{v}$$

λ = eigenvalue

$\underline{v}$ = eigenvector

λ_p 's are roots of characteristic polynomial

$$p(\lambda) = \det(\underline{\underline{S}} - \lambda \underline{\underline{I}}) = 0$$

For each λ_p we have one or more $\underline{v}_p$'s satisfying

$$(\underline{\underline{S}} - \lambda_p \underline{\underline{I}}) \underline{v}_p = \underline{0}$$

In continuum mechanics we are

mostly concerned with symmetric

tensors, e.g. stress. $\underline{\underline{\sigma}} = \underline{\underline{\sigma}}^T$

Eigen problem for symmetric tensors

- 1) All λ_p 's real
- 2) All λ_p 's are positive ($\leq$ sym. pos. def.)
- 3) All $\underline{v}_p$'s corresponding to distinct λ_p 's are orthogonal

$\underline{\underline{S}}$ is symmetric positive definite (spd) if $\underline{v} \cdot \underline{\underline{S}} \underline{v} > 0$ for all $\underline{v} \in \mathcal{V}$

by def. of eigenpair $\underline{\underline{S}} \underline{v} = \lambda \underline{v}$

$$\underline{v} \cdot \lambda \underline{v} = \lambda |\underline{v}|^2 > 0 \rightarrow \lambda > 0$$

For orthogonality consider two eigenpairs $(\lambda, \underline{v})$ and $(\omega, \underline{u})$ so that $\lambda \neq \omega$

$$\underline{\underline{S}} \underline{v} = \lambda \underline{v} \quad \text{and} \quad \underline{\underline{S}} \underline{u} = \omega \underline{u}$$

$$\begin{aligned} \text{Consider } \lambda (\underline{v} \cdot \underline{u}) &= \underline{\underline{S}} \underline{v} \cdot \underline{u} = \underline{v} \cdot \underline{\underline{S}}^T \underline{u} \quad (\underline{\underline{S}}^T = \underline{\underline{S}}) \\ &= \underline{v} \cdot \underline{\underline{S}} \underline{u} = \omega (\underline{v} \cdot \underline{u}) \end{aligned}$$

$$\lambda (\underline{v} \cdot \underline{u}) = \omega (\underline{v} \cdot \underline{u})$$

$$\text{since } \lambda \neq \omega \Rightarrow \underline{v} \cdot \underline{u} = 0$$

Spectral decomposition

If $\underline{\underline{S}} = \underline{\underline{S}}^T$ we can choose the set of eigenvectors as frame $\{\underline{\underline{e}}'_i\}$ so that

where

$$[\underline{\underline{S}}]' = \sum_{i=1}^3 \lambda_i (\underline{\underline{e}}'_i \otimes \underline{\underline{e}}'_i) \quad \underline{\underline{e}}'_i = \underline{\underline{v}}_i$$

$$[\underline{\underline{S}}]' = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \text{diagonal tensor}$$

To see this consider $[\underline{\underline{S}}]'[\underline{\underline{v}}_i]' = \lambda_i [\underline{\underline{v}}_i]'$ where $[\underline{\underline{v}}_p]' = \underline{\underline{e}}'_p$

In frame $\{\underline{\underline{e}}'_i\}$ we have $[\underline{\underline{I}}]' = \underline{\underline{e}}'_i \otimes \underline{\underline{e}}'_i$

$$\begin{aligned} [\underline{\underline{S}}]'[\underline{\underline{I}}]' &= [\underline{\underline{S}}]'(\underline{\underline{e}}'_i \otimes \underline{\underline{e}}'_i) = ([\underline{\underline{S}}]'\underline{\underline{e}}'_i) \otimes \underline{\underline{e}}'_i \quad \text{we } \underline{\underline{S}}\underline{\underline{e}}'_i = \lambda_i \underline{\underline{e}}'_i \\ &= \sum_{i=1}^3 ([\underline{\underline{S}}]'\underline{\underline{e}}'_i) \otimes \underline{\underline{e}}'_i = \sum_{i=1}^3 (\lambda_i \underline{\underline{e}}'_i) \otimes \underline{\underline{e}}'_i \\ &= \sum_{i=1}^3 \lambda_i (\underline{\underline{e}}'_i \otimes \underline{\underline{e}}'_i) \end{aligned}$$

where we have used associative properties

of dyadic product: $\underline{\underline{A}}(\underline{\underline{a}} \otimes \underline{\underline{b}}) = (\underline{\underline{A}}\underline{\underline{a}}) \otimes \underline{\underline{b}} \Rightarrow \text{HW3}$

$$\alpha(\underline{\underline{a}} \otimes \underline{\underline{b}}) = (\alpha \underline{\underline{a}}) \otimes \underline{\underline{b}}$$

Change of basis $\{\underline{e}_i\}$ to $\{\underline{e}'_i\}$

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principal frame

$$\underline{\underline{A}} = (\underline{e}_i \cdot \underline{e}'_j) \quad \underline{e}_i \otimes \underline{e}_j$$

$$= \begin{bmatrix} [\underline{e}'_1] & [\underline{e}'_2] & [\underline{e}'_3] \end{bmatrix} = \begin{bmatrix} [\underline{v}_1] & [\underline{v}_2] & [\underline{v}_3] \end{bmatrix} = \underline{\underline{V}}$$

↑ ↑ ↑
representation of eigenvectors $\underline{\underline{S}} \underline{v}_p = \lambda_p \underline{v}_p$ in $\{\underline{e}_i\}$

$$\Rightarrow [\underline{\underline{S}}] = [\underline{\underline{V}}] [\underline{\underline{S}}'] [\underline{\underline{V}}]^T$$

relation to diagonalization of symmetric matrices:

$$[\underline{\underline{S}}'] = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} = \underline{\underline{\Lambda}}$$

$$[\underline{\underline{V}}]^T = [\underline{\underline{V}}]^{-1} \quad \text{orthogonal}$$

in matrices $[\underline{\underline{S}}] = \underline{\underline{S}}$ always in $\{\underline{e}_i\}$

$$\underline{\underline{S}} = \underline{\underline{V}} \underline{\underline{\Lambda}} \underline{\underline{V}}^{-1}$$

$$\underline{\underline{\Lambda}} = \underline{\underline{V}}^{-1} \underline{\underline{S}} \underline{\underline{V}}$$

↑ ↑
diagonal full

$$\underline{\underline{S}} = \underline{\underline{S}}^T$$

$\underline{\underline{V}}$ = eigenvector matrix

$\underline{\underline{\Lambda}}$ = diagonal eigenvalue mat.

$[\underline{\sigma}]$ representation in $\{e_i\}$

$\Rightarrow$ eigenvector tensor = change of basis

$$\underline{V} = [\underline{v}_1] [\underline{v}_2] [\underline{v}_3] \equiv \underline{A}$$

$$\{e_i\} \longleftrightarrow \{e'_i\} = \{v_i\}$$

change of basis between $\{e_i\}$ and $\{e'_i\}$

$\{e'_i\} = \{v_i\}$ principal frame/eigenframe

To find representation of stress in NED

$\Rightarrow$ represent eigenvectors in NED to

form $\underline{A}$